Highlights

Volatility Spillovers Between Stablecoins and Foreign Exchange Markets: A Dynamic Analysis with TVP-VAR
稳定币与外汇市场之间的波动溢出：基于 **TVP-VAR** 的动态分析

Yi Zhang

- We combine the Diebold–Yilmaz spillover index with a Kalman-filter TVP-VAR on GARCH volatilities.
- 我们将 Diebold–Yilmaz 溢出指数与基于 Kalman 滤波的 TVP-VAR 相结合，作用于 GARCH 波动率。
- In normal times stablecoin volatility is largely idiosyncratic; stablecoins are not net transmitters on average.
- 常态时期稳定币波动在很大程度上是特质性的；平均而言稳定币并非波动的净输出者。
- Spillovers spike during the Terra/LUNA collapse and the SVB-induced USDC de-peg.
- 溢出在 Terra/LUNA 崩盘与 SVB 引发的 USDC 脱锚期间急剧上升。
- During crises stablecoins flip to large net transmitters of volatility.
- 危机期间稳定币转变为波动率的大规模净输出者。
- Emerging market currencies are relatively more exposed to stablecoin volatility.
- 新兴市场货币相对更容易暴露于稳定币波动。

Volatility Spillovers Between Stablecoins and Foreign Exchange Markets: A Dynamic Analysis with TVP-VAR
稳定币与外汇市场之间的波动溢出：基于 TVP-VAR 的动态分析

Yi Zhang[a,*]

[a]*School of Economics and Finance, 123 University Avenue, Beijing, 100084, , China*

Abstract

This paper examines the dynamic volatility spillovers between major USD-pegged stablecoins (USDT, USDC, and DAI) and the foreign exchange markets of 27 economies from January 2020 to December 2025. We estimate GARCH(1,1) conditional volatilities for all series and apply the Diebold–Yilmaz spillover index framework based on generalized forecast error variance decompositions, together with a time-varying parameter vector autoregression (TVP-VAR) estimated by Kalman filtering with forgetting factors. We document four main findings. First, in normal times stablecoin price volatility is largely idiosyncratic: on average, stablecoin shocks account for only about 1.6% of the forecast error variance of exchange-rate volatility, and USDT and USDC are net receivers of volatility within the full 31-variable system (net spillovers of −22.8% and −3.5%, respectively), while DAI is a modest net transmitter (+8.4%). Second, connectedness is strongly time-varying and crisis-driven: the rolling total spillover index peaks at 90% on 13 March 2023, and the TVP-VAR total connectedness index jumps from 24.1% to 52.9% around the SVB-induced USDC de-peg. Third, stablecoins flip from net receivers to large net transmitters during stablecoin-specific crises: the net spillover of USDT rises by 97.9 percentage points around the Terra/LUNA collapse of May 2022, and the net spillovers of USDC and DAI rise by 96.7 and 78.0 percentage points around the March 2023 SVB event. Fourth, emerging market currencies are relatively more exposed to stablecoin volatility than developed market currencies, both in group-level subsystems (net stablecoin spillovers of +0.84% in the emerging market system versus +0.36% in the developed market system) and in currency-level net receipts. Our results indicate that stablecoin–FX volatility linkages are episodic rather than structural, and that monitoring stablecoin stress is most valuable as an early-warning device during crypto-market crises, particularly for emerging economies.

本文考察 2020 年 1 月至 2025 年 12 月期间主要美元锚定稳定币（USDT、USDC、DAI）与 27 个经济体外汇市场之间的动态波动溢出。我们对所有序列估计 GARCH(1,1) 条件波动率，并基于广义预测误差方差分解应用 Diebold–Yilmaz 溢出指数框架，同时采用带遗忘因子的 Kalman 滤波估计时变参数向量自回归（TVP-VAR）模型。本文得到四个主要发现：第一，在常态时期，稳定币价格波动在很大程度上是特质性的——平均而言，稳定币冲击仅解释汇率波动预测误差方差的约 1.6%；在包含 31 个变量的全系统中，USDT 和 USDC 是波动率的净接收者（净溢出分别为 −22.8% 和 −3.5%），而 DAI 是温和的净输出者（+8.4%）。第二，连通性具有显著的时变性并由危机驱动：滚动总溢出指数在 2023 年 3 月 13 日达到 90% 的峰

*Corresponding author.
Email address: `yzhang@university.edu` (Yi Zhang)

值，TVP-VAR 总连通性指数在硅谷银行（SVB）引发的 USDC 脱锚事件前后从 24.1% 跃升至 52.9%。第三，在稳定币特有的危机期间，稳定币由净接收者转变为大规模净输出者：2022 年 5 月 Terra/LUNA 崩盘前后 USDT 净溢出上升 97.9 个百分点；2023 年 3 月 SVB 事件前后 USDC 和 DAI 的净溢出分别上升 96.7 和 78.0 个百分点。第四，新兴市场货币相对发达经济体货币更容易受到稳定币波动的影响，无论是子系统层面（新兴市场系统内稳定币净溢出为 +0.84%，发达市场系统为 +0.36%）还是逐币种净接收层面均如此。本文结果表明，稳定币与外汇之间的波动联动是事件性而非结构性的；监测稳定币压力作为加密市场危机期的预警工具最具价值，对新兴经济体尤其如此。

Keywords: Stablecoins, Volatility spillovers, TVP-VAR, Foreign exchange, Financial stability
稳定币, 波动溢出, TVP-VAR, 外汇市场, 金融稳定

---

1. Introduction / 引言

The rapid growth of USD-pegged stablecoins has transformed them from niche crypto trading tools into systemically important components of global financial markets. With a combined market capitalization exceeding $150 billion as of 2025, stablecoins facilitate trillions of dollars in transactions annually, serving as a bridge between traditional fiat currencies and crypto assets. As their role has expanded, so have concerns about their potential impact on financial stability, particularly in foreign exchange markets where stablecoins are increasingly used as informal dollar substitutes.

美元锚定稳定币的快速增长已使其从小众的加密交易工具转变为全球金融市场中具有系统重要性的组成部分。截至 2025 年，稳定币总市值超过 1500 亿美元，每年促成数万亿美元的交易，充当传统法定货币与加密资产之间的桥梁。随着其作用不断扩大，关于其对金融稳定潜在影响的担忧也在增加，尤其是在外汇市场——稳定币正日益被用作非正式的美元替代品。

This paper provides a comprehensive empirical analysis of volatility spillovers between stablecoin markets and traditional foreign exchange markets. We make three contributions. First, we estimate GARCH(1,1) conditional volatilities for three major stablecoins (USDT, USDC, DAI), 27 exchange rates against the US dollar, and the VIX, and we apply the Diebold–Yilmaz spillover index framework based on generalized forecast error variance decompositions (GFEVD) to quantify the magnitude and direction of volatility spillovers. This allows us to measure total spillovers, directional spillovers, and net pairwise spillovers between individual stablecoins and currencies.

本文对稳定币市场与传统外汇市场之间的波动溢出进行了全面的实证分析。我们有三方面贡献：第一，对三种主要稳定币（USDT、USDC、DAI）、27 种兑美元汇率以及 VIX 指数估计 GARCH(1,1) 条件波动率，并基于广义预测误差方差分解（GFEVD）应用 Diebold–Yilmaz 溢出指数框架，量化波动溢出的规模与方向，从而测度总溢出、方向性溢出以及单个稳定币与货币之间的成对净溢出。

Second, we estimate a time-varying parameter VAR (TVP-VAR) by Kalman filtering with forgetting factors, following the refined connectedness approach of Antonakakis et al. (2020), to examine how spillover dynamics evolved over a sample rich in stress events: the Terra/LUNA collapse in May 2022, the SVB-related USDC de-pegging in March 2023, and the emerging market currency turbulence of late 2024.

第二，我们遵循 Antonakakis et al. (2020) 的精炼连通性方法，采用带遗忘因子的 Kalman 滤波估计时变参数 VAR（TVP-VAR），以考察在富含压力事件的样本期内溢出动态如何演化——样本期涵盖 2022 年 5 月 Terra/LUNA 崩盘、2023 年 3 月 SVB 相关的 USDC 脱锚以及

2024 年末的新兴市场货币动荡。

Third, we analyze spillovers separately for USDT, USDC, and DAI and separately for developed and emerging market currency groups, because these stablecoins differ in user bases, collateral structures, and risk profiles, and because emerging market currencies are plausibly more exposed to stablecoin shocks.

第三，我们分别考察 USDT、USDC、DAI 以及发达与新兴市场货币组别的溢出，因为这些稳定币在用户基础、抵押结构和风险特征上存在差异，且新兴市场货币在理论上更易受到稳定币冲击的影响。

Our analysis yields four main findings, which qualify the prevailing narrative that stablecoins are a standing source of volatility for FX markets. First, in normal times stablecoin price volatility is largely idiosyncratic: on average, shocks to the three stablecoins together explain only about 1.6% of the forecast error variance of exchange-rate volatility, and within the full 31-variable system USDT and USDC are net receivers of volatility while DAI is a modest net transmitter. Hypotheses that stablecoins are permanent net transmitters are therefore rejected for the full sample on average. Second, connectedness is strongly time-varying and crisis-driven: the rolling total spillover index peaks at 90% on 13 March 2023, and the TVP-VAR total connectedness index jumps from 24.1% to 52.9% around the SVB event. Third, stablecoins flip from net receivers to large net transmitters during stablecoin-specific crises: USDT's net spillover rises by 97.9 percentage points around the Terra/LUNA collapse, and USDC's and DAI's net spillovers rise by 96.7 and 78.0 percentage points around the SVB event. Fourth, emerging market currencies are relatively more exposed to stablecoin volatility than developed market currencies.

本文的分析得出四个主要发现，对”稳定币是外汇市场持续性波动源”这一流行叙事进行了修正。第一，常态时期稳定币价格波动 largely 是特质性的：平均而言，三种稳定币的冲击合计仅解释汇率波动预测误差方差的约 1.6%；在 31 变量全系统中，USDT 与 USDC 是波动率的净接收者，DAI 为温和的净输出者。因此，”稳定币是永久净输出者”的假说在全样本平均意义上被拒绝。第二，连通性高度时变且由危机驱动：滚动总溢出指数在 2023 年 3 月 13 日达到 90% 的峰值，TVP-VAR 总连通性指数在 SVB 事件前后由 24.1% 跃升至 52.9%。第三，在稳定币特有的危机期间，稳定币由净接收者转变为大规模净输出者：Terra/LUNA 崩盘前后 USDT 净溢出上升 97.9 个百分点，SVB 事件前后 USDC 与 DAI 净溢出分别上升 96.7 和 78.0 个百分点。第四，新兴市场货币相对发达经济体货币更容易暴露于稳定币波动。

The paper proceeds as follows. Section 2 provides institutional background and reviews the literature. Section 3 describes the data. Section 4 outlines the methodology. Section 5 presents the Diebold–Yilmaz results. Section 6 presents the TVP-VAR results. Section 7 reports the event study. Section 8 conducts robustness checks. Section 9 concludes.

本文结构如下：第 2 节介绍制度背景并回顾文献；第 3 节描述数据；第 4 节阐述方法论；第 5 节报告 Diebold–Yilmaz 结果；第 6 节报告 TVP-VAR 结果；第 7 节报告事件研究；第 8 节进行稳健性检验；第 9 节总结全文。

2. Institutional Background and Literature Review / 制度背景与文献综述

2.1. The Stablecoin Ecosystem / 稳定币生态

The stablecoin ecosystem has evolved dramatically since the launch of USDT in 2014. Today, three major USD-pegged stablecoins dominate.

自 2014 年 USDT 问世以来，稳定币生态经历了剧烈演变。目前，三种主要的美元锚定稳定币占据主导地位。

USDT (Tether). As the oldest and largest stablecoin, USDT has a market capitalization of approximately $85 billion as of 2025. It is issued by Tether Limited and backed by a combination of cash, cash equivalents, commercial paper, and other assets. USDT dominates retail crypto trading and is widely used in emerging markets as an alternative to USD.

USDT（泰达币）. 作为历史最久、规模最大的稳定币，USDT 截至 2025 年市值约 850 亿美元，由 Tether Limited 发行，以现金、现金等价物、商业票据及其他资产组合作为储备支持。USDT 主导零售端加密交易，并在新兴市场被广泛用作美元替代品。

USDC (USD Coin). Launched in 2018, USDC has a market capitalization of approximately $48 billion. It is issued by Circle, a regulated financial institution, and backed by cash and short-dated US Treasury securities. USDC is favored by institutional investors due to its transparency and regulatory compliance.

USDC（*USD Coin*）. 于 2018 年推出，市值约 480 亿美元，由受监管的金融机构 Circle 发行，以现金和短期美国国债作为储备支持。凭借透明度与合规性，USDC 更受机构投资者青睐。

DAI. Launched in 2019, DAI is a decentralized stablecoin with a market capitalization of approximately $5 billion. Unlike USDT and USDC, DAI is maintained by the MakerDAO protocol and backed by crypto collateral. It is used primarily within the DeFi ecosystem.

DAI. 于 2019 年推出，是市值约 50 亿美元的去中心化稳定币。与 USDT、USDC 不同，DAI 由 MakerDAO 协议维护，以加密资产作为抵押，主要应用于 DeFi 生态。

2.2. Related Literature / 相关文献

Our paper is related to three strands of literature.
本文与三支文献相关。

Volatility Spillovers in FX Markets. A large literature examines volatility spillovers in foreign exchange markets. Early work by Engle et al. (1990) documents volatility spillovers across major currencies. Diebold and Yilmaz (2009, 2012, 2014) develop the spillover index framework that we use, and Antonakakis et al. (2020) extend it to time-varying parameter VARs. Several papers apply this framework to FX markets, including Antonakakis and Kizys (2015) and Greenwood-Nimmo et al. (2016). We contribute by examining spillovers between stablecoins and traditional FX markets.

外汇市场波动溢出. 大量文献考察外汇市场的波动溢出。Engle et al. (1990) 的早期工作记录了主要货币之间的波动溢出；Diebold and Yilmaz (2009, 2012, 2014) 发展了本文采用的溢出指数框架；Antonakakis et al. (2020) 将其推广到时变参数 VAR。Antonakakis and Kizys (2015)、Greenwood-Nimmo et al. (2016) 等将该框架应用于外汇市场。本文的贡献在于考察稳定币与传统外汇市场之间的溢出。

Stablecoins and Financial Stability. A growing literature examines the financial stability implications of stablecoins. Eichengreen (2022) argues that stablecoins could pose risks to financial stability through runs and disintermediation. Gorton and Zhang (2021) draw parallels between stablecoins and historical private money. We contribute by quantifying the time-varying volatility spillovers between stablecoins and FX markets.

稳定币与金融稳定. 越来越多的文献考察稳定币的金融稳定含义。Eichengreen (2022) 认为稳定币可能通过挤兑与脱媒对金融稳定构成风险；Gorton and Zhang (2021) 将稳定币与历史上的私人货币相类比。本文的贡献在于量化稳定币与外汇市场之间时变的波动溢出。

Crypto Volatility Spillovers. Several papers examine volatility spillovers within crypto markets. Borri (2019) documents spillovers between Bitcoin and other crypto assets. Antonakakis et al. (2019) examine connectedness in crypto markets. We contribute by examining spillovers between crypto markets (stablecoins) and traditional financial markets (FX).

加密市场波动溢出. 部分文献考察加密市场内部的波动溢出。Borri (2019) 记录了比特币与其他加密资产之间的溢出；Antonakakis et al. (2019) 考察加密市场的连通性。本文的贡献在于考察加密市场（稳定币）与传统金融市场（外汇）之间的溢出。

3. Data / 数据

Our analysis uses daily data from 8 January 2020 to 30 December 2025 (1,297 common trading days). We combine data from several sources.

本文使用 2020 年 1 月 8 日至 2025 年 12 月 30 日的日度数据（共 1,297 个共同交易日），数据来自多个来源。

Stablecoin Data. We obtain daily closing prices of USDT, USDC, and DAI against the US dollar from the Kraken exchange. For each stablecoin we compute daily log returns; peg deviations are small in normal times but widen sharply during de-peg events (e.g., USDC traded near $0.967 on Kraken around the SVB failure).

稳定币数据. 我们从 Kraken 交易所获取 USDT、USDC 与 DAI 兑美元的日度收盘价，并据此计算日度对数收益率。常态时期锚定偏离很小，但在脱锚事件中急剧扩大（例如 SVB 倒闭前后 USDC 在 Kraken 一度交易于 0.967 美元附近）。

Exchange Rate Data. We use daily spot exchange rates (local currency per USD) for 27 economies from a multi-source panel built from FRED, the ECB Statistical Data Warehouse, and Yahoo Finance. The developed market currencies are EUR, GBP, JPY, CAD, AUD, CHF, SEK, NOK, SGD, KRW, PLN; the emerging market currencies are CNY, MXN, BRL, INR, RUB, ZAR, TRY, ARS, COP, CLP, THB, MYR, PHP, IDR, NGN, SAR.

汇率数据. 我们使用 27 个经济体兑美元的日度即期汇率（本币/美元），来自由 FRED、欧洲央行统计数据仓库与 Yahoo Finance 构建的多来源面板。发达市场货币包括 EUR、GBP、JPY、CAD、AUD、CHF、SEK、NOK、SGD、KRW、PLN；新兴市场货币包括 CNY、MXN、BRL、INR、RUB、ZAR、TRY、ARS、COP、CLP、THB、MYR、PHP、IDR、NGN、SAR。

Global Controls. We include the VIX index (CBOE) as a global risk-sentiment control in the volatility system.

全球控制变量. 我们在波动率系统中纳入 VIX 指数（CBOE）作为全球风险偏好控制变量。

We compute daily volatility for each variable using a GARCH(1,1) model:

$$r_{i,t} = \mu_i + \epsilon_{i,t} \tag{1}$$

$$\epsilon_{i,t} = \sigma_{i,t}\eta_{i,t} \tag{2}$$

$$\sigma^2_{i,t} = \omega_i + \alpha_i\epsilon^2_{i,t-1} + \beta_i\sigma^2_{i,t-1} \tag{3}$$

Table 1: 波动率度量的描述性统计 / Summary Statistics: Volatility Measures

Variable / 变量	Mean / 均值	Std. Dev. / 标准差	Min / 最小值	Max / 最大值
Panel A: Stablecoin GARCH volatility (%) / 稳定币波动率				
USDT	0.043	0.034	0.011	0.270
USDC	0.022	0.063	0.006	0.922
DAI	0.107	0.236	0.014	2.558
Panel B: Developed market FX (%) / 发达市场汇率				
EUR	0.500	0.112	0.334	0.870
GBP	0.592	0.164	0.408	1.358
JPY	0.642	0.195	0.372	1.500
Average / 平均 (11)	0.607	0.112	0.465	1.103
Panel C: Emerging market FX (%) / 新兴市场汇率				
TRY	0.908	1.100	0.403	20.833
ARS	0.513	1.170	0.293	22.640
NGN	1.225	1.478	0.421	13.197
Average / 平均 (16)	0.739	0.184	0.506	2.101
Panel D: Global control / 全球控制变量				
VIX	8.221	2.682	6.412	37.831
Time period / 样本期	2020-01-08 – 2025-12-30			
Observations / 观测数	1,297 per series / 每序列			

where $r_{i,t}$ is the daily log return (in percent) for asset i, $\eta_{i,t}$ is iid standard normal, and $\sigma^2_{i,t}$ is the conditional variance. All 31 GARCH(1,1) models converge.

我们对每个变量使用 GARCH(1,1) 模型估计日度波动率，其中 $r_{i,t}$ 为资产 i 的日度对数收益率（百分比），$\eta_{i,t}$ 为独立同分布的标准正态扰动，$\sigma^2_{i,t}$ 为条件方差。全部 31 个 GARCH(1,1) 模型均收敛。

Table 1 presents summary statistics for our volatility measures, and Figure 1 plots the conditional volatility series.

表 1 给出了波动率度量的描述性统计，图 1 绘制了条件波动率序列。

The summary statistics show that stablecoin price volatility is an order of magnitude smaller than FX volatility in normal times: mean GARCH volatility is 0.043% for USDT and 0.022% for USDC, against 0.61% for the average developed market currency and 0.74% for the average emerging market currency. DAI is the most volatile stablecoin (mean 0.107%). Stablecoin volatility is, however, strongly episodic: maxima of 0.27% (USDT), 0.92% (USDC), and 2.56% (DAI) are reached around de-peg events. Emerging market FX volatility is higher than developed market FX volatility, with extreme values for ARS (22.6%), TRY (20.8%), and NGN (13.2%) reflecting currency crises.

描述性统计显示，常态时期稳定币价格波动比外汇波动低一个数量级：USDT 与 USDC 的平均 GARCH 波动率分别为 0.043% 和 0.022%，而发达市场货币平均为 0.61%，新兴市场货币平均为 0.74%；DAI 是波动最大的稳定币（均值 0.107%）。然而稳定币波动具有强烈的事件性：在脱锚事件前后，USDT、USDC、DAI 波动率最大值分别达到 0.27%、0.92%、2.56%。新兴市场汇率波动高于发达市场，ARS（22.6%）、TRY（20.8%）、NGN（13.2%）的极端值反映了货币危机。

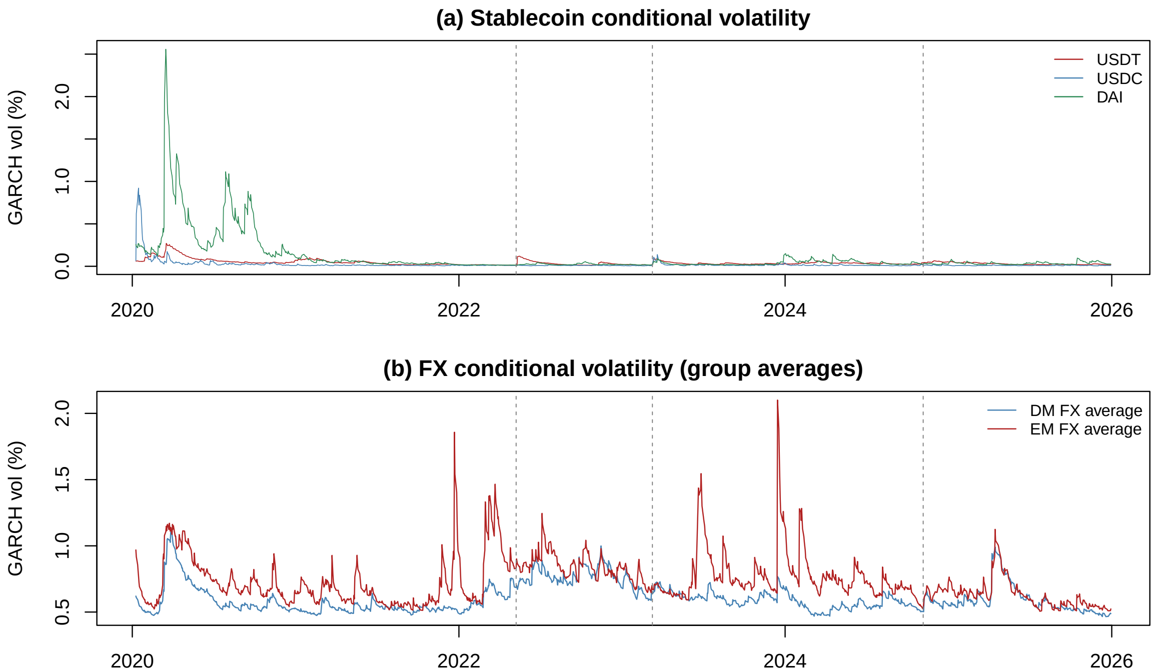

Figure 1: GARCH(1,1) 条件波动率 / GARCH(1,1) Conditional Volatility

Notes: Panel (a) plots GARCH(1,1) conditional volatilities of USDT, USDC, and DAI daily returns; Panel (b) plots the developed-market and emerging-market averages of FX conditional volatilities. Dashed lines mark the Terra/LUNA collapse (2022-05-09), the SVB/USDC event (2023-03-10), and the 2024 emerging market turbulence (2024-11-05).

注：(a) 子图为 USDT、USDC、DAI 日收益率的 GARCH(1,1) 条件波动率；(b) 子图为发达市场与新兴市场汇率条件波动率的组内平均。虚线分别标注 Terra/LUNA 崩盘（2022-05-09）、SVB/USDC 事件（2023-03-10）与 2024 年新兴市场动荡（2024-11-05）。

4. Methodology / 方法论

4.1. Diebold–Yilmaz Spillover Index / Diebold–Yilmaz 溢出指数

We use the Diebold–Yilmaz (DY) spillover index framework (Diebold and Yilmaz, 2009, 2012, 2014) to quantify volatility spillovers. The framework is based on variance decompositions from a vector autoregression (VAR).

我们采用 Diebold–Yilmaz（DY）溢出指数框架（Diebold and Yilmaz, 2009, 2012, 2014）量化波动溢出。该框架基于向量自回归（VAR）的方差分解。

Consider a VAR(p) model for our volatility series:

$$\mathbf{y}_t = \mathbf{c} + \sum_{k=1}^{p} \mathbf{A}_k \mathbf{y}_{t-k} + \boldsymbol{\epsilon}_t \tag{4}$$

where $\mathbf{y}_t = (\sigma_{1,t}, \sigma_{2,t}, \ldots, \sigma_{N,t})'$ is an $N \times 1$ vector of volatility measures, $\mathbf{c}$ is a vector of constants, $\mathbf{A}_k$ are coefficient matrices, and $\boldsymbol{\epsilon}_t \sim (0, \Sigma)$ is a vector of innovations.

考虑波动率序列的 VAR(p) 模型，其中 $\mathbf{y}_t = (\sigma_{1,t}, \ldots, \sigma_{N,t})'$ 为 $N \times 1$ 波动率向量，$\mathbf{c}$ 为常数向量，$\mathbf{A}_k$ 为系数矩阵，$\boldsymbol{\epsilon}_t \sim (0, \Sigma)$ 为新息向量。

The VAR can be rewritten in its moving average (MA) representation:

$$\mathbf{y}_t = \sum_{k=0}^{\infty} \mathbf{\Phi}_k \boldsymbol{\epsilon}_{t-k} \tag{5}$$

where $\mathbf{\Phi}_k$ are MA coefficient matrices computed recursively.

该 VAR 可改写为移动平均（MA）表示，其中 $\mathbf{\Phi}_k$ 为递归计算的 MA 系数矩阵。

We use the generalized forecast error variance decomposition (GFEVD) of Koop et al. (1996) and Pesaran and Shin (1998), which is invariant to variable ordering. The GFEVD measures the contribution of shocks to variable j to the forecast error variance of variable i at horizon H:

$$\theta_{ij}^{g}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (\mathbf{e}_i' \mathbf{\Phi}_h \Sigma \mathbf{e}_j)^2}{\sum_{h=0}^{H-1} (\mathbf{e}_i' \mathbf{\Phi}_h \Sigma \mathbf{\Phi}_h' \mathbf{e}_i)} \tag{6}$$

where σ_{jj} is the j-th diagonal element of Σ, and $\mathbf{e}_i$ is a selection vector with 1 in the i-th position and 0 otherwise.

我们采用 Koop et al. (1996) 与 Pesaran and Shin (1998) 的广义预测误差方差分解（GFEVD），其结果不依赖变量排序。GFEVD 测度变量 j 的冲击对变量 i 在 H 期预测误差方差的贡献，其中 σ_{jj} 为 Σ 的第 j 个对角元素，$\mathbf{e}_i$ 为选择向量。

We normalize the GFEVD so that each row sums to 1:

$$\tilde{\theta}_{ij}^{g}(H) = \frac{\theta_{ij}^{g}(H)}{\sum_{j=1}^{N} \theta_{ij}^{g}(H)} \tag{7}$$

我们对 GFEVD 进行行归一化，使每行之和为 1。

Using these normalized variance decompositions, we define the total spillover index:

$$S^{g}(H) = \frac{\sum_{i,j=1, i \neq j}^{N} \tilde{\theta}_{ij}^{g}(H)}{N} \times 100 \tag{8}$$

This measures the contribution of spillovers across all variables to the total forecast error variance.

基于归一化方差分解，总溢出指数定义为所有跨变量溢出对总预测误差方差贡献的平均。

We also define directional spillovers. The directional spillover from all other variables to variable i (“FROM”) and from variable i to all other variables (“TO”) are:

$$S_{i \leftarrow \bullet}^{g}(H) = \sum_{j=1, j \neq i}^{N} \tilde{\theta}_{ij}^{g}(H) \times 100, \qquad S_{\bullet \leftarrow i}^{g}(H) = \sum_{j=1, j \neq i}^{N} \tilde{\theta}_{ji}^{g}(H) \times 100 \tag{9}$$

方向性溢出定义为：变量 i 接收自所有其他变量的溢出（FROM）与输出至所有其他变量的溢出（TO）。

The net directional spillover and the net pairwise spillover are:

$$S_{i}^{g}(H) = S_{\bullet \leftarrow i}^{g}(H) - S_{i \leftarrow \bullet}^{g}(H), \qquad S_{ij}^{g}(H) = \left(\tilde{\theta}_{ji}^{g}(H) - \tilde{\theta}_{ij}^{g}(H) \right) \times 100 \tag{10}$$

净方向性溢出与成对净溢出分别按上述公式定义。

4.2. TVP-VAR Model / TVP-VAR 模型

To examine time variation in spillover dynamics, we estimate a time-varying parameter VAR:

$$\mathbf{y}_t = \mathbf{c}_t + \sum_{k=1}^{p} \mathbf{A}_{k,t}\mathbf{y}_{t-k} + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim (0, \Sigma_t) \tag{11}$$

where the coefficients $\mathbf{c}_t$, $\mathbf{A}_{k,t}$ and the covariance matrix Σ_t are time-varying.

为考察溢出动态的时变性，我们估计系数 $\mathbf{c}_t$、$\mathbf{A}_{k,t}$ 与协方差矩阵 Σ_t 均随时间变化的 TVP-VAR 模型。

We let $\alpha_t = \text{vec}([\mathbf{c}_t, \mathbf{A}_{1,t}, \ldots, \mathbf{A}_{p,t}])$ follow a random walk:

$$\alpha_t = \alpha_{t-1} + \nu_t, \quad \nu_t \sim (0, Q) \tag{12}$$

令 $\alpha_t = \text{vec}([\mathbf{c}_t, \mathbf{A}_{1,t}, \ldots, \mathbf{A}_{p,t}])$ 服从随机游走。

Following Koop and Korobilis (2013) and the refined connectedness implementation of Antonakakis et al. (2020), we estimate the model with a Kalman filter using forgetting factors: the state covariance is inflated by $1/\kappa_1$ each period ($\kappa_1 = 0.99$), and the error covariance follows an exponentially weighted moving average with decay $\kappa_2 = 0.96$. Time-varying GFEVD and total connectedness indices TCI_t are computed from the filtered coefficients and covariances at each date. As a complement, we also compute rolling-window DY spillover indices with a 120-day window.

遵循 Koop and Korobilis (2013) 以及 Antonakakis et al. (2020) 的精炼连通性实现，我们使用带遗忘因子的 Kalman 滤波估计模型：状态协方差每期按 $1/\kappa_1$ 放大（$\kappa_1 = 0.99$），误差协方差服从衰减系数为 $\kappa_2 = 0.96$ 的指数加权移动平均。在每个时点，由滤波得到的系数与协方差计算时变 GFEVD 与总连通性指数 TCI_t。作为补充，我们还计算 120 日滚动窗口的 DY 溢出指数。

All connectedness measures are computed with the `ConnectednessApproach` R package (Gabauer, 2022), which implements the generalized-FEVD framework of Diebold and Yilmaz (2012, 2014), rolling-window time connectedness, and the TVP-VAR connectedness of Antonakakis et al. (2020). GARCH models are estimated with `rugarch`.

所有连通性度量均使用 R 包 `ConnectednessApproach`（Gabauer, 2022）计算，该包实现了 Diebold and Yilmaz (2012, 2014) 的广义 FEVD 框架、滚动窗口时变连通性以及 Antonakakis et al. (2020) 的 TVP-VAR 连通性；GARCH 模型使用 `rugarch` 估计。

5. Main Results: Diebold–Yilmaz Spillover Index / 主要结果：Diebold–Yilmaz 溢出指数

We estimate the baseline VAR on the full volatility panel: 3 stablecoins, 27 FX series, and the VIX (N = 31). We use a lag length of 2, selected by the AIC (HQ and BIC select 1; results are robust, see Section 8), and a forecast horizon of 10 days.

我们在完整波动率面板上估计基准 VAR：3 种稳定币、27 个汇率序列与 VIX（N = 31）。滞后阶数取 2（由 AIC 选定；HQ 与 BIC 选择 1；结果对此稳健，见第 8 节），预测期域为 10 天。

Table 2: 全样本溢出表 / Spillover Table: Full Sample

	Own 自身	USDT	USDC	DAI	FX (27)	VIX	FROM 接收	TO 输出	NET 净溢出
USDT	58.14	–	0.72	20.51	17.14	3.47	41.86	19.08	−22.78
USDC	94.21	1.08	–	2.13	2.42	0.16	5.79	2.26	−3.53
DAI	57.16	3.15	0.43	–	37.86	1.41	42.84	51.19	+8.35
FX avg. / 汇率均值	38.63	0.51	0.04	1.02	58.10	1.70	61.37	61.82	+0.44
VIX	54.97	1.05	0.00	0.96	43.03	–	45.03	50.99	+5.96

Notes: Row-normalized generalized FEVD (%) from a VAR(2) on 31 GARCH volatility series, horizon 10 days. "Own" is the share of a variable's forecast error variance (FEV) explained by its own shocks; FROM (TO) is the sum of FEV shares received from (transmitted to) all other variables; NET = TO − FROM. The FX row averages the 27 currency rows; the FX column sums the 27 currency columns. Total spillover index: 57.83%. Average share of FX FEV explained by stablecoin shocks: USDT 0.51%, USDC 0.04%, DAI 1.02% (total 1.57%).

注：基于 31 个 GARCH 波动率序列 VAR(2) 的行归一化广义 FEVD（%），预测期域 10 天。" 自身" 为变量预测误差方差（FEV）中由自身冲击解释的份额；FROM（TO）为接收自（输出至）所有其他变量的 FEV 份额之和；NET = TO − FROM。FX 行为 27 个货币行的平均，FX 列为 27 个货币列的加总。总溢出指数为 57.83%；稳定币冲击平均解释汇率 FEV 的份额：USDT 0.51%、USDC 0.04%、DAI 1.02%（合计 1.57%）。

5.1. Full-Sample Spillover Table / 全样本溢出表

Table 2 presents the full-sample spillover table, aggregated to the three stablecoins, the average FX series, and the VIX.

表 2 给出全样本溢出表，聚合为三种稳定币、平均汇率序列与 VIX。

The full-sample results qualify the hypothesis that stablecoins are standing net transmitters of volatility. The total spillover index averages 57.83%, indicating substantial connectedness within the joint system, but this connectedness is dominated by FX–FX and VIX links: on average, shocks to the three stablecoins jointly explain only about 1.57% of the forecast error variance of exchange-rate volatility (USDT 0.51%, USDC 0.04%, DAI 1.02%). Within the full system, USDT and USDC are net receivers of volatility (net spillovers of −22.78% and −3.53%), while DAI is a modest net transmitter (+8.35%). Stablecoin volatility is largely idiosyncratic in normal times—USDC's own shocks explain 94.2% of its forecast error variance—and where stablecoins do interact, they interact mainly with each other: 20.5% of USDT's FEV is explained by DAI shocks.

全样本结果对" 稳定币是持续性波动净输出者" 的假说构成了修正。总溢出指数平均为 57.83%，表明联合系统内部连通性可观，但该连通性主要由外汇—外汇及 VIX 关联主导：平均而言，三种稳定币的冲击合计仅解释汇率波动预测误差方差的约 1.57%（USDT 0.51%、USDC 0.04%、DAI 1.02%）。在全系统中，USDT 与 USDC 为波动率净接收者（净溢出分别为 −22.78% 与 −3.53%），DAI 为温和净输出者（+8.35%）。常态时期稳定币波动在很大程度上是特质性的——USDC 自身冲击解释其 FEV 的 94.2%——而稳定币之间的相互作用主要发生在稳定币内部：USDT 的 FEV 中有 20.5% 由 DAI 冲击解释。

5.2. Time-Varying Total Spillovers: Rolling Estimates / 时变总溢出：滚动估计

Figure 2 plots the total spillover index computed with a 120-day rolling window on a core 10-variable system (USDT, USDC, DAI, EUR, JPY, GBP, CNY, TRY, ARS, VIX). Total spillovers vary substantially over time: the index averages 42.5%, dips below 30% during quiet periods (early 2021, early 2022), and spikes during stress episodes—most notably to 90.0% on 13 March 2023, the Monday after the SVB failure when USDC de-pegged. Secondary spikes occur around

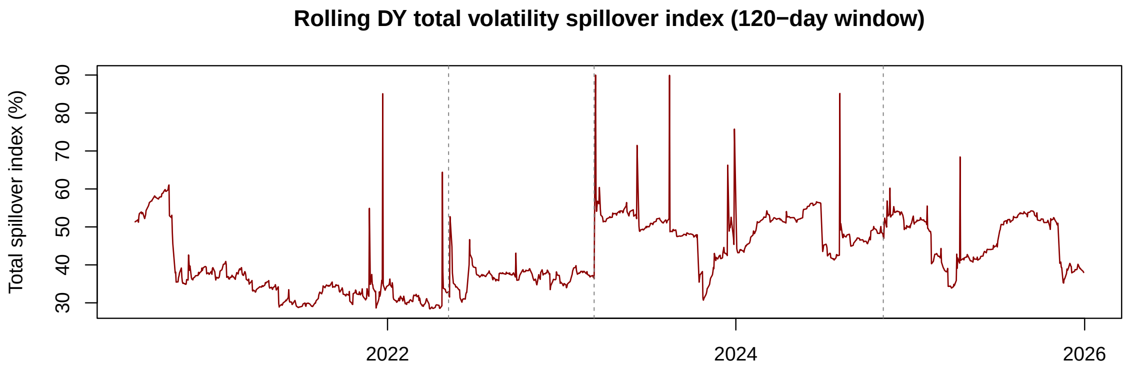

Figure 2: 滚动总溢出指数 / Rolling Total Spillover Index

Notes: Total spillover index from a VAR(1) on the core 10-variable volatility system, 120-day rolling window, horizon 10 days. Dashed lines mark the three stress events.
注：核心 10 变量波动率系统 VAR(1)、120 日滚动窗口、10 天期域下的总溢出指数。虚线标注三次压力事件。

the Terra/LUNA collapse (May 2022), the March 2020 COVID turmoil (visible at the left edge of the rolling window), and several emerging market stress episodes in 2024–2025.

图 2 绘制了基于核心 10 变量系统（USDT、USDC、DAI、EUR、JPY、GBP、CNY、TRY、ARS、VIX）的 120 日滚动窗口总溢出指数。总溢出随时间大幅波动：指数平均为 42.5%，在平静时期（2021 年初、2022 年初）降至 30% 以下，并在压力时期急剧飙升——最显著的是在 2023 年 3 月 13 日（SVB 倒闭后 USDC 脱锚的周一）达到 90.0%。次级峰值出现在 Terra/LUNA 崩盘（2022 年 5 月）、2020 年 3 月新冠疫情动荡（滚动窗口左端可见）以及 2024–2025 年的几次新兴市场压力时期。

5.3. Directional Spillovers / 方向性溢出

Figure 3 plots directional spillovers between the stablecoin group and the FX group within the core system. Spillovers from FX to stablecoins (blue) exceed spillovers from stablecoins to FX (red) in most periods, consistent with the full-sample finding that stablecoins are on average net receivers. The gap narrows or reverses sharply around stablecoin-specific stress events—the Terra/LUNA collapse and the SVB de-peg—when stablecoin-to-FX spillovers temporarily rise.

图 3 绘制了核心系统内稳定币组与外汇组之间的方向性溢出。在多数时期，外汇对稳定币的溢出（蓝线）高于稳定币对外汇的溢出（红线），与全样本中稳定币平均为净接收者的发现一致。在稳定币特有的压力事件——Terra/LUNA 崩盘与 SVB 脱锚——前后，两者差距急剧收窄甚至反转，稳定币对外汇的溢出暂时上升。

5.4. Net Spillovers by Stablecoin / 分稳定币的净溢出

Figure 4 plots rolling net spillovers for each stablecoin. The figure makes the episodic nature of stablecoin transmission clear: USDT's net spillover spikes sharply around the Terra/LUNA collapse (May 2022); USDC's net spillover spikes during the SVB event (March 2023); DAI shows positive spikes around both crypto crises. Outside these windows, net spillovers hover around or below zero.

图 4 绘制了每种稳定币的滚动净溢出。该图清晰展示了稳定币溢出的事件性特征：USDT 净溢出在 Terra/LUNA 崩盘（2022 年 5 月）前后急剧飙升；USDC 净溢出在 SVB 事件（2023

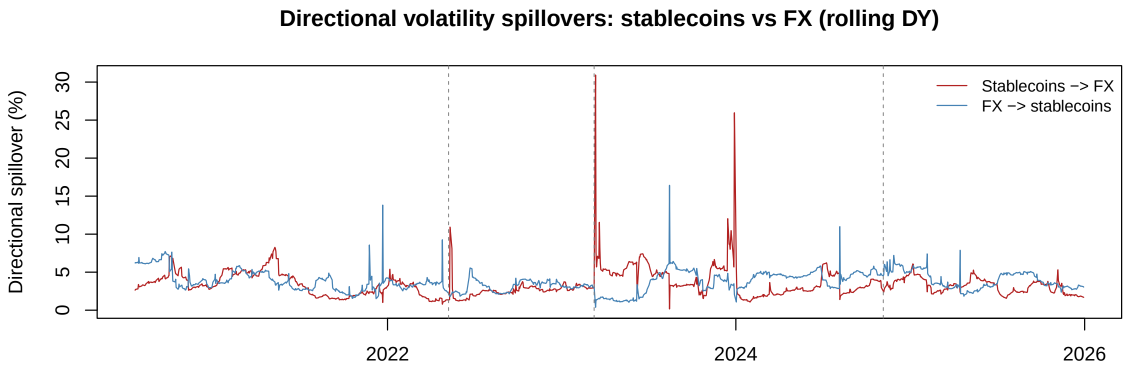

Figure 3: 方向性溢出：稳定币与外汇 / Directional Spillovers: Stablecoins vs FX

Notes: Average share of forecast error variance (per variable, %) transmitted from the stablecoin group to the FX group (red) and from the FX group to the stablecoin group (blue), rolling DY estimates.
注：稳定币组向外汇组（红线）与外汇组向稳定币组（蓝线）传递的平均预测误差方差份额（按变量平均，%），滚动 DY 估计。

年 3 月）期间飙升；DAI 在两次加密危机前后均出现正向脉冲。在这些窗口之外，净溢出大体徘徊在零附近或以下。

5.5. Pairwise Spillovers / 成对溢出

Table 3 presents the top 10 pairwise net spillovers from stablecoins to individual currencies (full sample), and Figure 5 shows the net pairwise spillover matrix for the core system. Pairwise spillovers from stablecoins to FX are small but not uniformly distributed: the largest go to emerging market currencies (COP, CLP, BRL, SAR, RUB, IDR) together with JPY, KRW, and CAD. DAI and USDT account for most of the top pairwise links.

表 3 给出稳定币对单个货币的十大成对净溢出（全样本），图 5 展示核心系统的成对净溢出矩阵。稳定币对外汇的成对溢出较小但分布不均：最大的溢出流向新兴市场货币（COP、CLP、BRL、SAR、RUB、IDR）以及 JPY、KRW、CAD；DAI 与 USDT 占据了主要成对关联。

5.6. Developed vs Emerging Markets / 发达与新兴市场对比

We estimate separate subsystems for the developed market (DM) and emerging market (EM) currency groups, each containing the three stablecoins, the group's currencies, and the VIX. Table 4 summarizes the results, and Figure 6 plots the full-sample net spillover received by each currency from the stablecoin group.

我们分别对发达市场（DM）与新兴市场（EM）货币组估计子系统，每个子系统包含三种稳定币、该组货币与 VIX。表 4 汇总结果，图 6 绘制全样本中每种货币接收自稳定币组的净溢出。

Three regularities emerge. First, stablecoins are net transmitters within both group subsystems, and more so in the EM subsystem (+0.84%) than in the DM subsystem (+0.36%). Second, the stablecoin-to-FX channel is larger in the EM subsystem (0.83% of FEV) than in the DM subsystem (0.76%), even though the EM subsystem is less internally connected overall (TCI 31.5% vs 57.4%). Third, in the full system, the average EM currency receives a less negative net spillover from stablecoins (−0.46%) than the average DM currency (−0.69%); the largest net

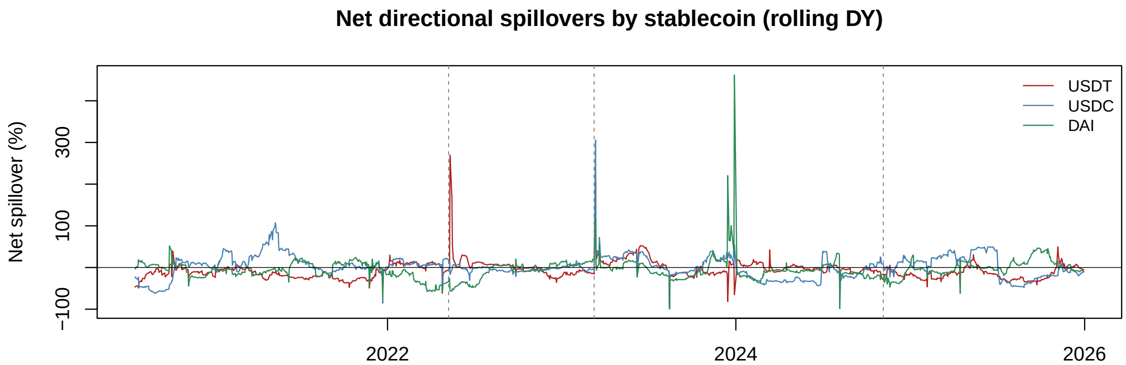

Figure 4: 分稳定币的净溢出 / Net Spillovers by Stablecoin

Notes: Rolling net directional spillovers (%) for each stablecoin in the core system. Positive values indicate a net transmitter of volatility.
注：核心系统中各稳定币的滚动净方向性溢出（%）。正值表示波动率净输出者。

receivers from stablecoins include SAR, BRL, CLP (EM) and KRW (DM). Taken together, the evidence supports the hypothesis that emerging market currencies are relatively more exposed to stablecoin volatility, although the magnitudes are small in normal times.

三点规律浮现：第一，稳定币在两个子系统内均为净输出者，且在新兴市场子系统（+0.84%）中强于发达市场子系统（+0.36%）。第二，稳定币对外汇的溢出通道在新兴市场子系统（占 FEV 的 0.83%）大于发达市场子系统（0.76%），尽管新兴市场子系统整体内部连通性更低（TCI 31.5% 对 57.4%）。第三，在全系统中，新兴市场货币平均接收自稳定币的净溢出（−0.46%）负值小于发达市场货币（−0.69%）；接收稳定币净溢出最多的货币包括 SAR、BRL、CLP（新兴市场）与 KRW（发达市场）。综合来看，证据支持新兴市场货币相对更易暴露于稳定币波动的假说，尽管常态时期量级很小。

6. TVP-VAR Results: Time-Varying Spillovers / TVP-VAR 结果：时变溢出

We now present results from the TVP-VAR model on the core 10-variable system, which traces spillover dynamics more flexibly than the rolling window approach.

下面给出核心 10 变量系统 TVP-VAR 模型的结果，相比滚动窗口方法，它能更灵活地刻画溢出动态。

6.1. Time-Varying Total Spillover Index / 时变总溢出指数

Figure 7 plots the TVP-VAR total connectedness index against the rolling DY index. The two measures agree closely on the location of spikes—the COVID turmoil, the Terra/LUNA collapse, the SVB event, and the 2024–2025 emerging market stress episodes—while the TVP-VAR index is smoother and uses information from the entire sample. The TVP-VAR index averages 34.3% and peaks at 73.6% on 15 December 2023.

图 7 将 TVP-VAR 总连通性指数与滚动 DY 指数并列绘制。两种度量在峰值位置上高度一致——新冠动荡、Terra/LUNA 崩盘、SVB 事件以及 2024–2025 年新兴市场压力时期——而 TVP-VAR 指数更平滑且利用了全样本信息。TVP-VAR 指数平均为 34.3%，于 2023 年 12 月 15 日达到 73.6% 的峰值。

Table 3: 稳定币对外汇的十大成对净溢出 / Top 10 Pairwise Net Spillovers: Stablecoins to FX

Pair / 货币对	Net spillover (%) / 净溢出	Share of positive (%) / 占正向比重
DAI → COP	1.41	14.5
USDT → CLP	1.20	12.3
USDT → JPY	0.76	7.8
DAI → BRL	0.66	6.8
USDT → KRW	0.66	6.7
USDT → CAD	0.54	5.6
USDT → SAR	0.54	5.5
DAI → RUB	0.49	5.1
USDC → SAR	0.44	4.6
USDT → IDR	0.37	3.7
Top 10 total / 前十大合计	7.07	72.6
All positive pairs / 全部正溢出对合计	9.74	100.0

Notes: Net pairwise directional spillovers (%) from the full-sample GFEVD (VAR(2), N = 31, H = 10). Positive values indicate that the currency is a net receiver from the stablecoin.
注：全样本 GFEVD（VAR(2)，N = 31，H = 10）的成对净方向性溢出（%）。正值表示该货币为稳定币溢出的净接收者。

6.2. Time-Varying Net Spillovers / 时变净溢出

Figure 8 plots TVP-VAR net spillovers for each stablecoin. The crisis-driven regime switching is striking: USDT's net spillover jumps from deeply negative to about +56% around the Terra/LUNA collapse; USDC's and DAI's net spillovers jump to about +73% and +59% around the SVB event. Between crises, stablecoin net spillovers decay back toward zero or negative values. This pattern—long quiet periods punctuated by sharp transmission spikes—is the central time-series stylized fact of stablecoin–FX connectedness.

图 8 绘制各稳定币的 TVP-VAR 净溢出。危机驱动的状态切换十分显著：USDT 净溢出在 Terra/LUNA 崩盘前后由深度负值跃升至约 +56%；USDC 与 DAI 净溢出在 SVB 事件前后跃升至约 +73% 与 +59%。危机之间，稳定币净溢出逐渐衰减回零或负值。"长期平静期与尖锐溢出脉冲交替"是稳定币—外汇连通性的核心时序典型事实。

Figure 9 compares the time-varying net pairwise spillover from USDT to the Turkish lira with that to the euro. The USDT → TRY spillover is systematically larger and more persistent than the USDT → EUR spillover, and it spikes around the 2024 emerging market turbulence, consistent with the greater exposure of emerging market currencies documented in Section 5.

图 9 比较了 USDT 对土耳其里拉与对欧元的时变成对净溢出。USDT → TRY 溢出一贯大于且持续于 USDT → EUR 溢出，并在 2024 年新兴市场动荡前后飙升，与第 5 节中新兴市场货币更易受影响的发现一致。

7. Event Study Analysis / 事件研究分析

We conduct an event study around three stress events: the Terra/LUNA collapse (9 May 2022), the SVB-induced USDC de-peg (10 March 2023), and the emerging market currency turbulence (5 November 2024). For each event we compare the average spillover index in the 10 days before and after the event date.

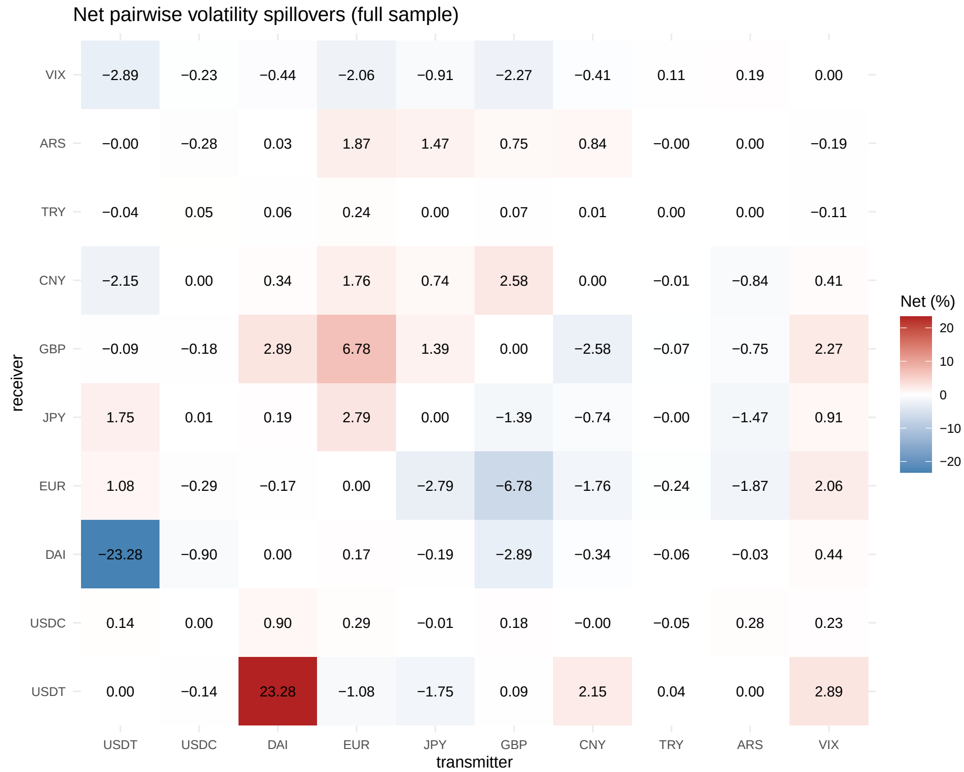

Figure 5: 成对净溢出热力图 / Net Pairwise Spillover Heatmap

Notes: Net pairwise spillovers (%) within the core 10-variable system, full sample. Rows are receivers, columns are transmitters; positive (red) entries indicate net transmission from the column variable to the row variable. 注：核心 10 变量系统全样本成对净溢出（%）。行为接收者，列为输出者；正值（红色）表示从列变量到行变量的净输出。

我们围绕三次压力事件进行事件研究：Terra/LUNA 崩盘（2022 年 5 月 9 日）、SVB 引发的 USDC 脱锚（2023 年 3 月 10 日）以及新兴市场货币动荡（2024 年 11 月 5 日）。对每个事件，我们比较事件日前后各 10 天的平均溢出指数。

Table 5 and Figure 10 show that spillovers increase around all three events, but with a clear hierarchy. The SVB/USDC event dominates: the TVP-VAR index rises by 28.8 points (24.1% to 52.9%), the rolling index by 20.7 points (37.1% to 57.8%, peaking at 90.0%), and the net spillovers of USDC and DAI rise by 96.7 and 78.0 points as the de-peg propagated through the system. The Terra/LUNA collapse operates mainly through USDT, whose net spillover rises by 97.9 points while the aggregate indices rise moderately. The 2024 emerging market turbulence is associated with smaller aggregate spillover increases but a visible rise in the USDT-to-TRY pairwise spillover (Figure 9), consistent with an FX-originated rather than stablecoin-originated episode.

表 5 与图 10 显示溢出在三次事件前后均上升，但层次分明。SVB/USDC 事件影响最大：TVP-VAR 指数上升 28.8 点（24.1% 至 52.9%），滚动指数上升 20.7 点（37.1% 至 57.8%，峰值 90.0%），USDC 与 DAI 的净溢出分别上升 96.7 与 78.0 点，脱锚冲击经由系统扩散。Terra/LUNA 崩盘主要通过 USDT 传导，其净溢出上升 97.9 点，而总量指数温和上升。2024 年新兴市场动荡对应的总量溢出升幅较小，但 USDT 对 TRY 的成对溢出明显上升（图 9），符合该事件源于外汇端而非稳定币端的特征。

Table 4: 发达与新兴市场子系统 / Developed vs Emerging Market Subsystems

	DM system / 发达市场	EM system / 新兴市场
Total spillover index (%) / 总溢出指数	57.40	31.51
SC → FX (avg FEV share, %) / 稳 → 汇	0.762	0.828
FX → SC (avg FEV share, %) / 汇 → 稳	0.623	0.706
Net SC → FX / 净溢出	+0.140	+0.121
Mean net spillover of SCs (%) / 稳定币平均净溢出	+0.360	+0.841
Mean net received per FX (%) / 每币种平均净接收	−0.687	−0.460

Notes: Group subsystems contain the three stablecoins, the group currencies (11 DM / 16 EM), and the VIX; VAR(2), H = 10. The last row is computed from the full 31-variable system as the average per-currency net spillover received from the stablecoin group.

注：组内子系统包含三种稳定币、该组货币（11 个发达市场 / 16 个新兴市场）与 VIX；VAR(2)，H = 10。最后一行基于 31 变量全系统，为每种货币接收自稳定币组的平均净溢出。

8. Robustness Checks / 稳健性检验

Table 6 reports the full-sample total spillover index and stablecoin net spillovers under alternative specifications.

表 6 报告了不同设定下全样本总溢出指数与稳定币净溢出。

The qualitative conclusions are robust. The total spillover index remains in the 52–61% range across volatility models (GARCH vs EGARCH), lag lengths (1–3), and forecast horizons (5–20 days). USDT remains a net receiver on average in all full-sample specifications, and DAI remains a (weak) net transmitter in most. Excluding the crisis windows leaves the full-sample picture essentially unchanged (TCI 57.6%), confirming that the average results are not driven solely by the three events. The subsample split reveals an important time trend: in the pre-2022 subsample all three stablecoins are strong net receivers (net spillovers of −34.6%, −48.4%, and −32.7%), whereas post-2022 their net positions move substantially toward zero (−4.2%, −0.1%, −18.6%), indicating that stablecoin–FX linkages have strengthened as the stablecoin market has grown. Figure A.11 in the appendix shows that rolling total spillover dynamics are nearly identical under GARCH and EGARCH volatilities.

定性结论保持稳健。在不同波动率模型（GARCH 与 EGARCH）、滞后阶数（1–3）与预测期域（5–20 天）下，总溢出指数保持在 52–61% 区间；USDT 在所有全样本设定下平均均为净接收者，DAI 在多数设定下为（弱）净输出者。剔除危机窗口后全样本图景基本不变（TCI 57.6%），表明平均结果并非仅由三次事件驱动。子样本划分揭示了重要的时间趋势：2022 年前子样本中三种稳定币均为强净接收者（净溢出分别为 −34.6%、−48.4%、−32.7%），而 2022 年后其净头寸大幅向零收敛（−4.2%、−0.1%、−18.6%），表明随着稳定币市场扩张，稳定币与外汇的联动已经增强。附录图 A.11 显示，GARCH 与 EGARCH 波动率下滚动总溢出动态几乎一致。

9. Conclusion / 结论

This paper provides a comprehensive analysis of volatility spillovers between stablecoin markets and the foreign exchange markets of 27 economies from 2020 to 2025, using GARCH(1,1) volatilities, the Diebold–Yilmaz spillover index, and a Kalman-filter TVP-VAR.

本文使用 GARCH(1,1) 波动率、Diebold–Yilmaz 溢出指数与 Kalman 滤波 TVP-VAR，全面分析了 2020–2025 年稳定币市场与 27 个经济体外汇市场之间的波动溢出。

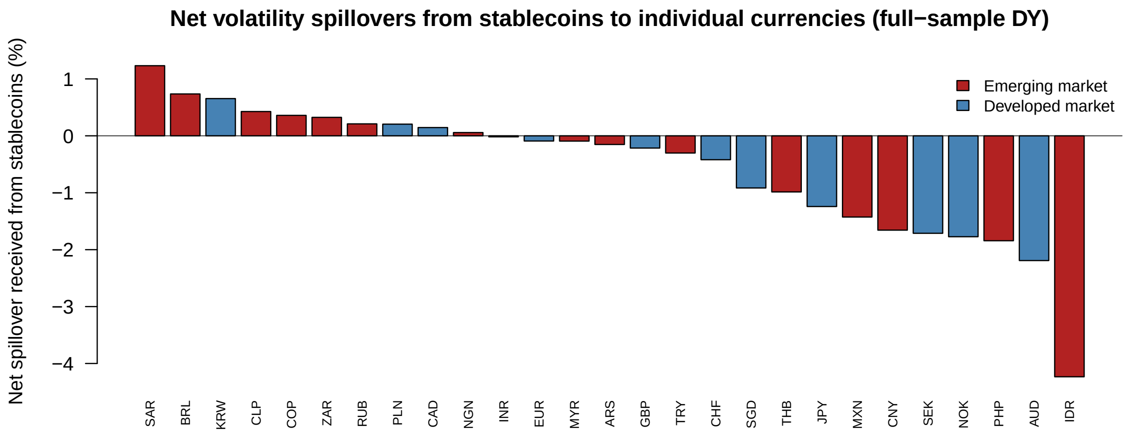

Figure 6: 各货币接收自稳定币的净溢出 / Net Spillover Received from Stablecoins by Currency

Notes: Full-sample net spillover (%) received by each currency from the stablecoin group (received minus transmitted), full 31-variable system. Red bars: emerging markets; blue bars: developed markets.
注：31 变量全系统中每种货币接收自稳定币组的全样本净溢出（%，接收减去输出）。红色柱为新兴市场，蓝色柱为发达市场。

Our findings qualify the prevailing narrative. On average, stablecoin price volatility is largely idiosyncratic, and stablecoins are not standing net transmitters of volatility to FX markets: stablecoin shocks explain only about 1.6% of FX volatility forecast error variance, and USDT and USDC are net receivers within the full system. Stablecoin–FX connectedness is instead episodic and crisis-driven: the total spillover index peaks at 90% around the SVB-induced USDC de-peg, and during stablecoin-specific crises stablecoins flip to large net transmitters—USDT around the Terra/LUNA collapse, USDC and DAI around the SVB event. Emerging market currencies are relatively more exposed than developed market currencies, and the stablecoin–FX linkage has strengthened over time.

本文的发现对流行叙事构成了修正。平均而言，稳定币价格波动 largely 是特质性的，稳定币并非外汇市场波动的持续性净输出者：稳定币冲击仅解释汇率波动预测误差方差的约 1.6%，USDT 与 USDC 在全系统中为净接收者。稳定币与外汇的连通性本质上是事件性、危机驱动的：总溢出指数在 SVB 引发的 USDC 脱锚前后达到 90% 的峰值；在稳定币特有的危机期间，稳定币转变为大规模净输出者——Terra/LUNA 崩盘时的 USDT、SVB 事件时的 USDC 与 DAI。新兴市场货币相对发达市场货币更易受影响，且稳定币与外汇的联动随时间增强。

These results carry a precise policy message. For central banks and regulators—particularly in emerging markets—stablecoin monitoring is most valuable as a crisis early-warning device rather than as a standing source of FX volatility risk: the information content of stablecoin stress concentrates in short windows around de-peg events. For international organizations, the cross-border amplification documented here during the March 2023 episode supports coordinated disclosure and reserve-quality standards for stablecoin issuers. Future research could examine intraday spillovers around de-peg announcements, the role of stablecoin flows (rather than prices) as transmission channels, and how central bank digital currencies would alter these dynamics.

上述结果传递了明确的政策含义。对各国央行与监管机构——尤其是新兴市场——而言，监测稳定币的最大价值在于充当危机预警工具，而非作为持续的汇率波动风险来源：稳定币压

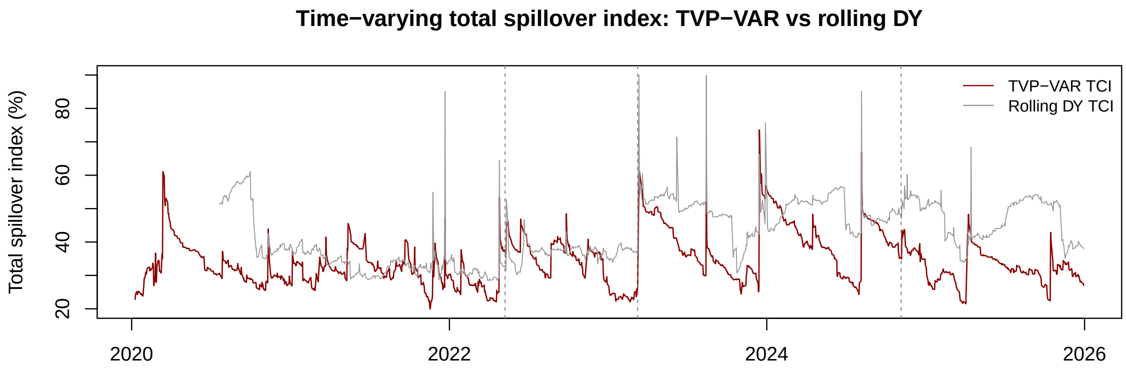

Figure 7: 时变总溢出指数（TVP-VAR 与滚动 DY）/ Time-Varying Total Spillover Index (TVP-VAR vs Rolling DY)

Notes: TVP-VAR total connectedness index (red) and rolling DY total spillover index (grey), core 10-variable system. Dashed lines mark the three stress events.
注： TVP-VAR 总连通性指数（红线）与滚动 DY 总溢出指数（灰线），核心 10 变量系统。虚线标注三次压力事件。

力的信息含量集中于脱锚事件前后的短窗口内。对国际组织而言，本文记录的 2023 年 3 月事件中的跨境放大效应支持对稳定币发行方实施协调的信息披露与储备质量标准。未来研究可考察脱锚公告前后的日内溢出、以稳定币流量（而非价格）作为传导渠道的作用，以及央行数字货币将如何改变这些动态。

References / 参考文献

Antonakakis, N., and Kizys, R. (2015). Dynamic spillovers between commodity and currency markets. International Review of Financial Analysis, 41, 303-319.

Antonakakis, N., Chatziantoniou, I., and Gabauer, D. (2019). Cryptocurrency market contagion: Market uncertainty, market complexity, and dynamic spillovers. Research in International Business and Finance, 48, 37-48.

Antonakakis, N., Chatziantoniou, I., and Gabauer, D. (2020). Refined measures of dynamic connectedness based on time-varying parameter vector autoregressions. Journal of Risk and Financial Management, 13(4), 84.

Borri, N. (2019). Conditional tail-risk in cryptocurrency markets. Journal of Empirical Finance, 50, 1-19.

Diebold, F. X., and Yilmaz, K. (2009). Measuring financial asset return and volatility spillovers, with application to global equity markets. Economic Journal, 119(534), 158-171.

Diebold, F. X., and Yilmaz, K. (2012). Better to give than to receive: Predictive directional measurement of volatility spillovers. International Journal of Forecasting, 28(1), 57-66.

Diebold, F. X., and Yilmaz, K. (2014). On the network topology of variance decompositions: Measuring the connectedness of financial firms. Journal of Econometrics, 182(1), 119-134.

Eichengreen, B. (2022). Stablecoins: Economic and regulatory issues. NBER Working Paper, 29713.

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. Econometrica, 50(4), 987-1007.

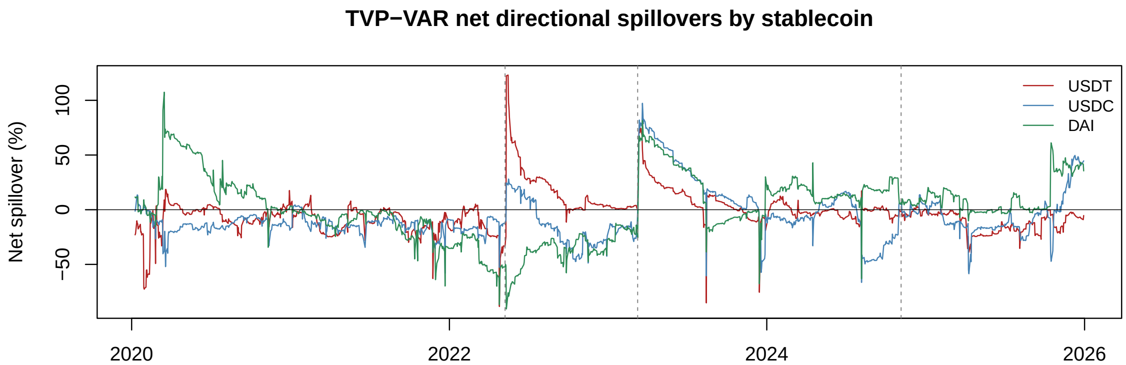

Figure 8: TVP-VAR 分稳定币时变净溢出 / TVP-VAR Net Spillovers by Stablecoin

Notes: TVP-VAR net directional spillovers (%) for USDT, USDC, and DAI in the core system. Positive values indicate a net transmitter.
注：核心系统中 USDT、USDC、DAI 的 TVP-VAR 净方向性溢出（%）。正值表示净输出者。

Engle, R. F., Ito, T., and Lin, W. L. (1990). Meteor showers or heat waves? Heteroskedastic intra-daily volatility in the foreign exchange market. Econometrica, 58(3), 525-542.

Gabauer, D. (2022). ConnectednessApproach: The estimation of connectedness measures. R package version 1.0.4. https://gabauerdavid.github.io/ConnectednessApproach/

Ghalanos, A. (2024). rugarch: Univariate GARCH models. R package.

Gorton, G., and Zhang, J. (2021). Stablecoins: Stability mechanisms and risks. NBER Working Paper, 28890.

Greenwood-Nimmo, M., Nguyen, V. H., and Rafferty, B. (2016). Risk and return spillovers among the G10 currencies. Journal of Financial Markets, 27, 42-59.

Koop, G., and Korobilis, D. (2013). Large time-varying parameter VARs. Journal of Econometrics, 177(2), 185-198.

Koop, G., Pesaran, M. H., and Potter, S. M. (1996). Impulse response analysis in nonlinear multivariate models. Journal of Econometrics, 74(1), 119-147.

Nakajima, J. (2011). Time-varying parameter VAR model with stochastic volatility: An overview of methodology and empirical applications. Monetary and Economic Studies, 29, 107-142.

Pesaran, H. H., and Shin, Y. (1998). Generalized impulse response analysis in linear multivariate models. Economics Letters, 58(1), 17-29.

Primiceri, G. E. (2005). Time varying structural vector autoregressions and monetary policy. Review of Economic Studies, 72(3), 821-852.

Appendix A. Additional Robustness Results / 附加稳健性结果

Figure A.11 compares rolling total spillover indices computed from GARCH(1,1) and EGARCH(1,1) conditional volatilities. The two series are nearly identical, confirming that the results are not sensitive to the choice of volatility model.

图 A.11 比较了基于 GARCH(1,1) 与 EGARCH(1,1) 条件波动率的滚动总溢出指数。两条序列几乎一致，证实结果对波动率模型的选择不敏感。

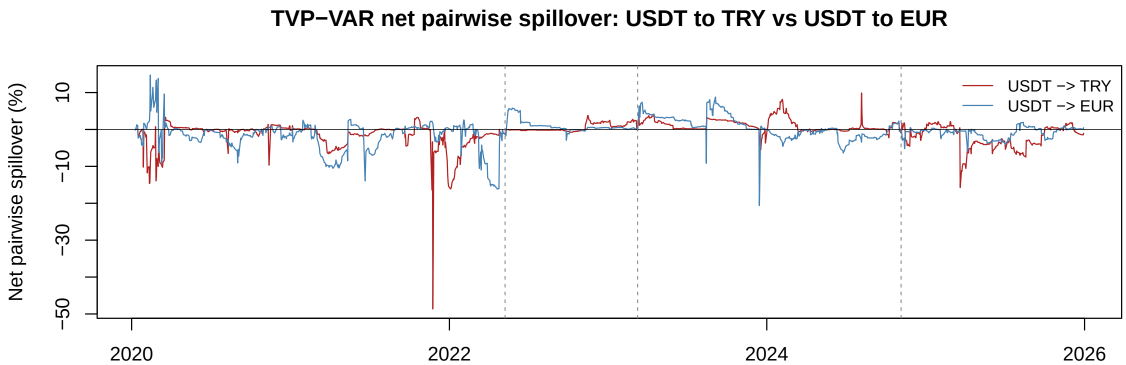

Figure 9: 时变成对净溢出：USDT 对 TRY 与 EUR / Time-Varying Net Pairwise Spillovers: USDT to TRY vs EUR

Notes: TVP-VAR net pairwise directional spillovers (%) from USDT to TRY (red) and to EUR (blue). Positive values indicate net transmission from USDT to the currency.
注：USDT 对 TRY（红线）与对 EUR（蓝线）的 TVP-VAR 成对净方向性溢出（%）。正值表示由 USDT 向该货币的净输出。

Appendix B. TVP-VAR Estimation Details / TVP-VAR 估计细节

The TVP-VAR is estimated by Kalman filtering with forgetting factors (Koop and Korobilis, 2013), as implemented in the TVP-VAR module of `ConnectednessApproach` (Gabauer, 2022), which follows Antonakakis et al. (2020). The state vector contains the intercept and the VAR(1) coefficients of the 10-variable system (100 coefficients); the state covariance is inflated by the factor $1/\kappa_1$ with $\kappa_1 = 0.99$ at each prediction step, and the error covariance matrix follows an exponentially weighted moving average of outer products of residuals with decay $\kappa_2 = 0.96$. The filter is initialized with a Minnesota-type prior centred on a random walk. Time-varying GFEVD matrices are computed from the filtered coefficients and covariances at each date with a 10-day horizon.

TVP-VAR 采用带遗忘因子的 Kalman 滤波估计（Koop and Korobilis, 2013），实现于 `ConnectednessApproach`（Gabauer, 2022）的 TVP-VAR 模块，遵循 Antonakakis et al. (2020)。状态向量包含 10 变量系统的截距与 VAR(1) 系数（共 100 个系数）；状态协方差在每个预测步按 $1/\kappa_1$（$\kappa_1 = 0.99$）放大；误差协方差矩阵服从残差外积的指数加权移动平均，衰减系数 $\kappa_2 = 0.96$。滤波器以集中于随机游走的 Minnesota 型先验初始化。时变 GFEVD 矩阵在每个时点由滤波系数与协方差按 10 天期域计算。

Appendix C. Event Dates / 事件日期

Our event study uses the following dates:
事件研究使用以下日期：

- Terra/LUNA collapse: May 9, 2022 (when UST de-pegged significantly).
- Terra/LUNA 崩盘：2022 年 5 月 9 日（UST 显著脱锚之日）。
- SVB/USDC event: March 10, 2023 (when SVB failed and USDC de-pegged).

Table 5: 压力事件前后的事件研究 / Event Study Around Stress Events

	Terra/LUNA 2022-05-09	SVB/USDC 2023-03-10	EM turbulence 2024-11-05
TVP-VAR total connectedness (%) / 总连通性指数			
Pre-event (10-day avg) / 事件前	36.68	24.08	37.76
Post-event (10-day avg) / 事件后	40.92	52.85	39.88
Change / 变化	+4.25	+28.78	+2.12
Rolling DY total spillover (%) / 滚动总溢出指数			
Pre-event / 事件前	35.49	37.06	48.84
Post-event / 事件后	37.22	57.75	52.58
Change / 变化	+1.73	+20.69	+3.73
TVP-VAR net spillover (%) / 净溢出			
USDT pre → post / 前 → 后	−41.42 → +56.43	+2.88 → +55.07	−4.66 → −4.75
change / 变化	+97.85	+52.18	−0.08
USDC pre → post	−18.63 → +13.58	−23.89 → +72.85	−13.84 → −1.89
change / 变化	+32.21	+96.74	+11.95
DAI pre → post	−60.35 → −68.86	−18.95 → +59.01	+21.25 → +6.52
change / 变化	−8.51	+77.96	−14.72
Peak in $[-20, +40]$ days / 窗口内峰值	64.41 / 53.39	90.00 / 61.13	60.22 / 43.75

Notes: Pre- and post-event levels are 10-day averages around the event date; the last row reports the peak of the rolling DY / TVP-VAR total spillover index within $[-20, +40]$ calendar days. The changes are descriptive; no inference is attached because the spillover indices are highly persistent.

注：事件前后水平为事件日前后各 10 天的平均值；最后一行为滚动 DY / TVP-VAR 总溢出指数在事件日前后 $[-20, +40]$ 个日历日内的峰值。变化值为描述性统计，由于溢出指数高度持续，未附加统计推断。

- SVB/USDC 事件：2023 年 3 月 10 日（SVB 倒闭、USDC 脱锚之日）。
- 2024 emerging market turbulence: November 5, 2024 (start of a period of severe EM currency pressure).
- 2024 年新兴市场动荡：2024 年 11 月 5 日（新兴市场货币严重承压期的起点）。

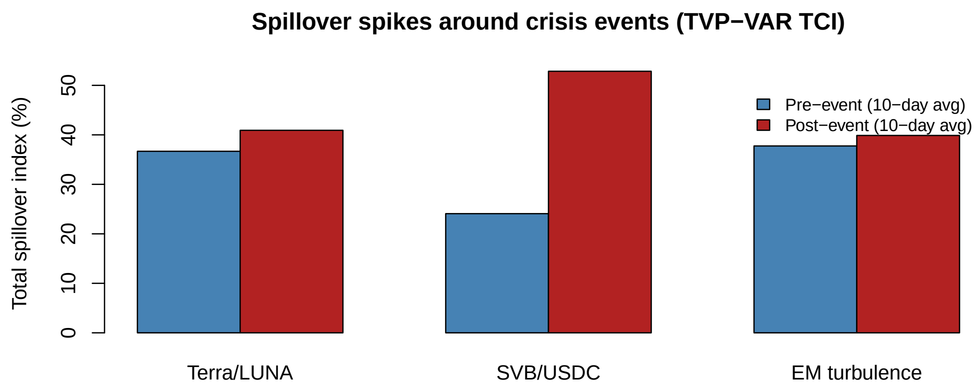

Figure 10: 危机事件前后的溢出峰值 / Spillover Spikes Around Crisis Events

Notes: Pre- and post-event 10-day averages of the TVP-VAR total connectedness index around each stress event.
注：各压力事件前后 TVP-VAR 总连通性指数的 10 天平均值。

Table 6: 稳健性检验 / Robustness Checks

Specification / 设定	TCI (%)	Net USDT	Net USDC	Net DAI
Baseline (GARCH, VAR(2), H = 10) / 基准	57.83	−22.78	−3.53	+8.35
EGARCH(1,1) volatility / EGARCH 波动率	52.20	−12.43	−18.17	+9.68
VAR lag 1 / 滞后 1 阶	56.76	−25.48	−1.99	+5.82
VAR lag 3 / 滞后 3 阶	59.48	−25.52	−11.96	+5.67
Horizon 5 days / 期域 5 天	54.83	−24.07	−1.36	+8.60
Horizon 20 days / 期域 20 天	61.45	−12.23	−4.42	+0.77
Pre-2022 subsample / 2022 年前子样本	69.17	−34.57	−48.39	−32.65
Post-2022 subsample / 2022 年后子样本	60.91	−4.17	−0.14	−18.56
Excluding crisis windows / 剔除危机窗口	57.64	−18.68	−2.99	+7.09

Notes: Full 31-variable system. Crisis windows exclude 15 days before to 30 days after each of the three stress events. Net spillovers in %.
注： 31 变量全系统。剔除危机窗口指剔除三次压力事件各自前 15 天至后 30 天的区间。净溢出单位为%。

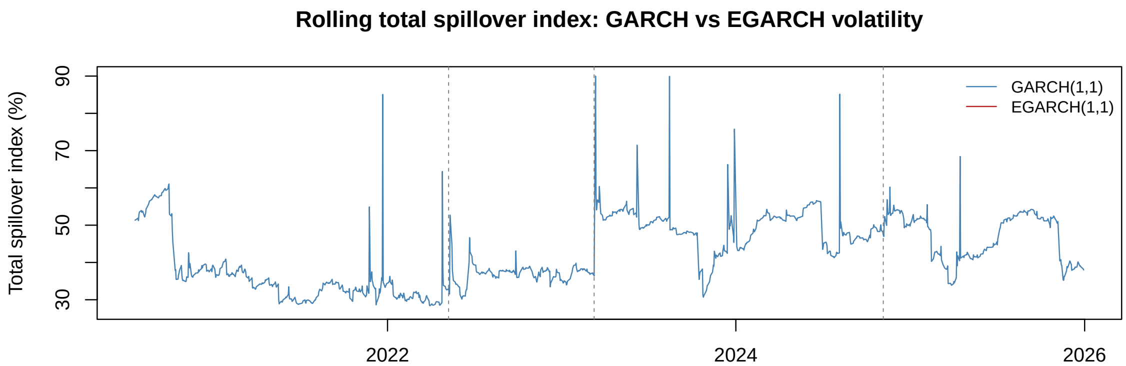

Figure A.11: 滚动总溢出指数：GARCH 与 EGARCH / Rolling Total Spillover Index: GARCH vs EGARCH

Notes: Rolling (120-day) total spillover index of the core 10-variable system, computed from GARCH(1,1) (blue) and EGARCH(1,1) (red) conditional volatilities.
注：核心 10 变量系统的 120 日滚动总溢出指数，分别基于 GARCH(1,1)（蓝线）与 EGARCH(1,1)（红线）条件波动率计算。